E-commerce Retail Sales Analysis

Problem Statement

E-commerce businesses generate large amounts of sales data through customer orders, product purchases, payment transactions, and marketing channels. Without proper analysis, it can be difficult to identify top-performing products, understand customer purchasing behaviour, monitor profitability, and evaluate sales performance across different regions.

The objective of this project is to analyze an e-commerce retail dataset using **Python, SQL, and Power BI** to identify meaningful business insights and support data-driven decision-making.

Key Areas of Analysis

- Evaluating total revenue, profit, and order performance.
- Identifying top-selling and most profitable products.
- Analyzing sales trends across months and years.
- Comparing revenue across product categories and states.
- Understanding customer purchasing and repeat purchase behaviour.
- Analyzing payment methods and customer acquisition channels.
- Identifying opportunities to improve sales and profitability.

Expected Outcome

An interactive Power BI dashboard that presents key performance indicators (KPIs), sales trends, product performance, and customer insights to help business stakeholders make informed decisions.